

ALBENI FALLS DAM SPILLWAY GATE REPLACEMENT

January 29-30, 2025

Presenters:

Daisy Douglass, NWS PM

David Kloewer, Tech Lead

Jeannette Patton, NWS Contract Specialist



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TOPICS

- Government Contracting Resource Introductions
- Albeni Falls Dam Basic Facts
- Albeni Falls Dam Current Status
- Overview of Replacement Scope
- Overview of Conceptual Design
- How you can stay informed?





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CONTRACTING INFORMATION



- **Contracting Officer:** Robert Gonzalez, robert.l.gonzalez@usace.army.mil
- **Contract Specialist:** Jeannette Patton, jeannette.k.patton@usace.army.mil
- **North American Industry Classification System (NAICS):** 332312 – Fabricated Structural Metal Manufacturing.
- **Product Service Code (PSC):** 9640 – Iron and Steel Primary and Semifinished Products
- Competitive, firm-fixed-price type contract to be procured in accordance with the Federal Acquisition Regulation FAR Part 15 (Contracting by Negotiation), using the Tradeoff process.
- Proposals will be evaluated based on source selection criteria the technical team has deemed relevant and important for confidence in success of the contract. The proposal deemed best value to the government will be selected (not lowest price, technically acceptable).
- 3 Gates will be awarded. Estimated duration of the project is 720 calendar days from issuance of the Notice to Proceed. Award will have options for potential other gates.



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USACE NWS OFFICE OF SMALL BUSINESS PROGRAMS



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USACE NWS SB Homepage:

<https://www.nws.usace.army.mil/Business-With-Us/Small-Business/>



Idaho APEX Accelerator

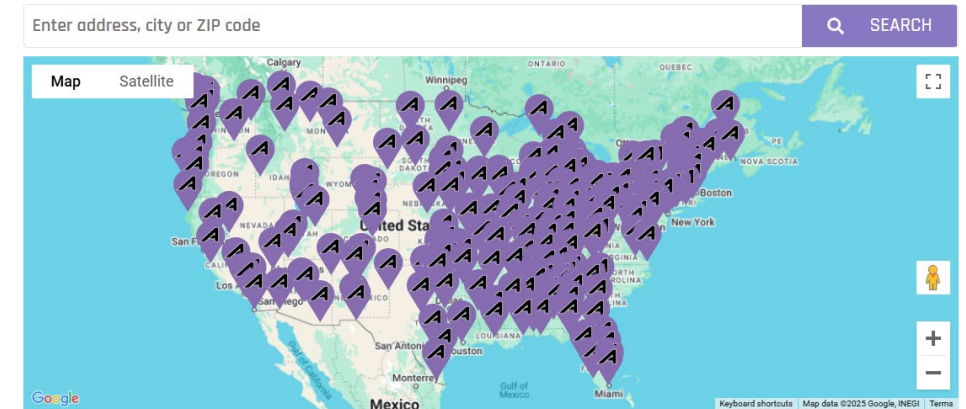
<https://www.idahoapexaccelerator.com/>

Matt Van Wagenen
State Director
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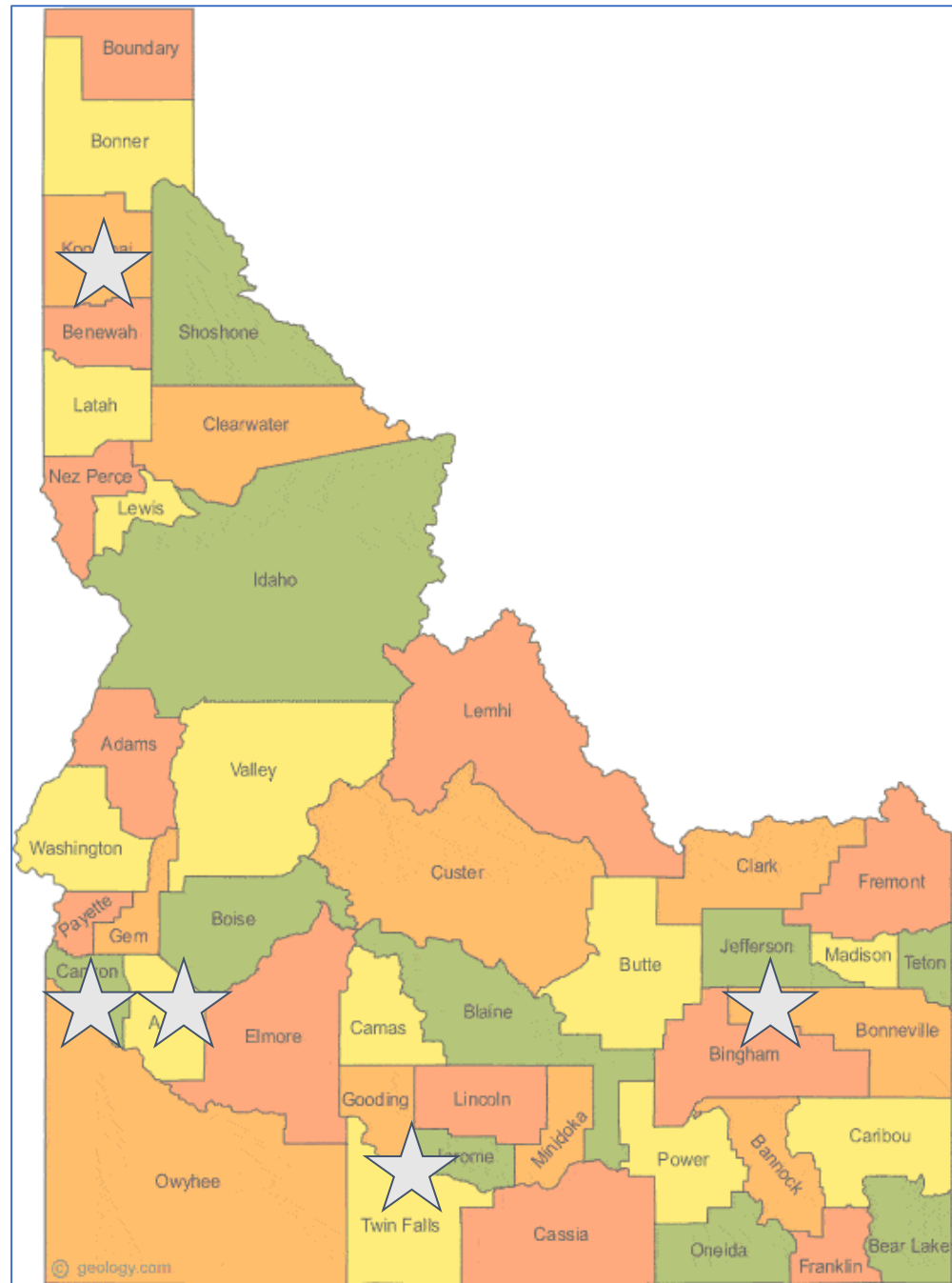
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OUR SERVICES



CORE (Individual)

- One-on-One Counseling
- SAM.gov Registration
- SBA Profile in the DSBS
- SBA Certifications
- Solicitation Review
- Marketing/Market Research
- Capability Statements

TRAINING (Group)

- Matchmaking Events
- Industry Days
- Regional / Statewide Training Events
- Small Business Liaison Office Collaboration
- Resource Partners

ADVANCED

- Cybersecurity/CMMC
- Gov't-led Innovation
- DoD & SBA Mentor/Protégé
- Foreign Ownership, Control, Influence (FOCI)
- Key Technology Areas



QUESTIONS

Thank you!

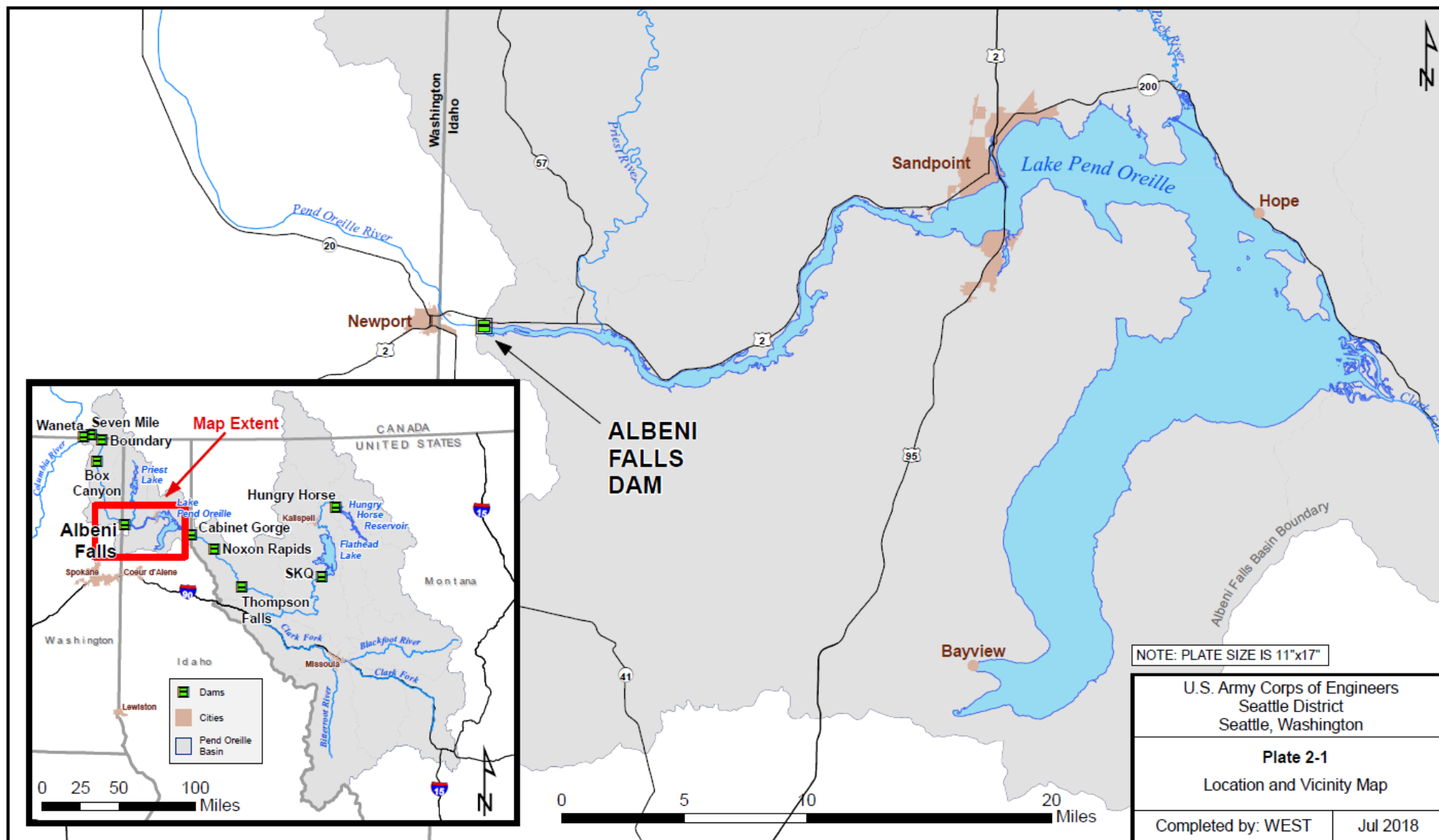




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ALBENI FALLS DAM AND LAKE PEND OREILLE





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ALBENI FALLS DAM: BASIC FACTS

Construction was completed in 1955.

The Dam consists of:

- **The Powerhouse** - passes water through three turbines to generate power and regulates lake level (fall drawdown and summer pool).
- **The Spillway** - passes water through 10 spillway gates to release water above powerhouse capacity (typically only used during spring runoff, not used for fall drawdown).
- **Storage** – the lake is generally operated between 2051 feet and 2062.5 feet, unless there are flood conditions. This equals about 1.0 maf.





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ALBENI FALLS DAM: BASIC DETAILS

- Authorized purposes: Hydropower, Flood Risk Management, Navigation, Recreation, and Fish and Wildlife Conservation (ENS).
- Powerhouse has 3 Kaplan generating units.
- Hydropower projects are 97.5% BPA funded.
- Dam, Spillway, FRM and Fish and Wildlife Conservation are joint funded between BPA and appropriation.
- Recreation and ENS value to the nation:
 - Provides public access to over 4,000 acres of public land, 10 boat ramps, 139 campsites and 23 recreation/access areas.
 - Approx. \$25M of visitor spending within 30 miles of the lake. *(Corps only, does not include other public or private entities).*





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DISCOVERY OF SPILLWAY GATE DEFECTS

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- Construction contract awarded SEPT 2022 for rehabilitation of 3 gates with 8 gates as options.
- Army Corps of Engineers initiated physical work on a gate rehabilitation contract in June 2023.
- The rehabilitation contract's purpose was to repair cracked welds identified during gate inspections and to re-paint the gate.
- As part of this rehabilitation contract:
 - Removed Gate number 3, put our spare gate in place.
 - Removed paint and completed in-depth gate inspection.
- During the gate rehab, steel defects were discovered in April 2024.



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ALBENI FALLS DAM DEFECT LOCATIONS

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- Base metal defects (holes and cracks in the steel)
- Should failure occur, the gate could fail and lead to crane operator injury and/or damage to the crane and bridge as well as loss of control of lake level.

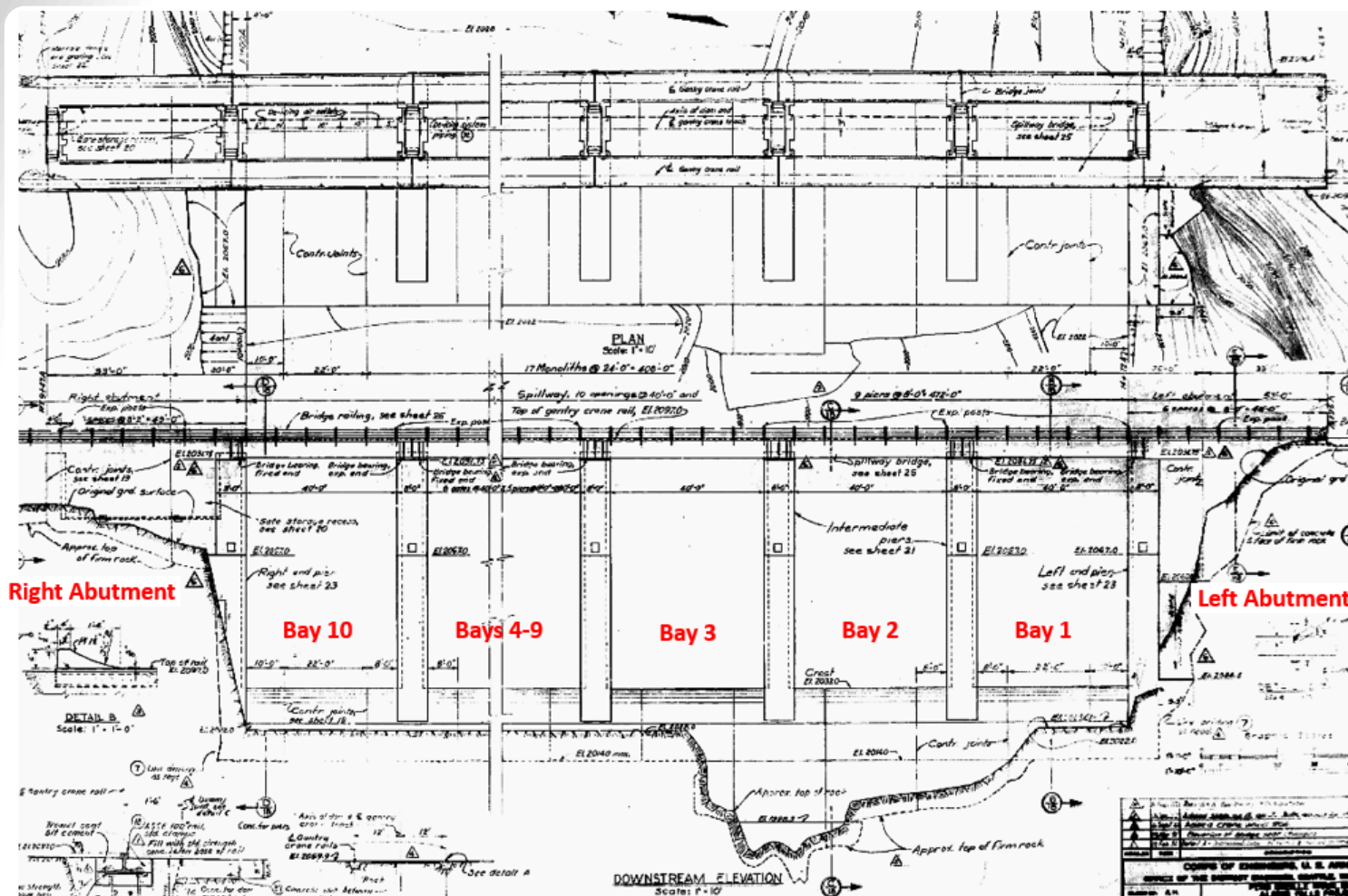




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SPILLWAY OVERVIEW

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Feature	Elevation (ft NGVD29)	Probable Head** (ft NGVD29)
Summer Pool* (at Hope)	2062.5-ft	32.5
Winter Pool* (at Hope)	2051.0-ft	21
Spillway Crest	2033.0-ft	3
Maximum Normal Regulated Pool* (at Hope)	2062.5-ft	32.5
Top of Gate(s)	2065.0-ft	35
Top of Dam (Spillway Bridge deck-roadway elevation)	2097.0-ft	
Pool of Record pre-dam* (1948, at Hope)	2071.6-ft	
Pool of Record post dam construction* (1997, at Hope)	2065.74-ft	

* Note that Lake elevations are measured at the USGS gage at Hope, Idaho, and not at the dam. Each lake elevation at Hope can result in multiple forebay elevations at the dam, depending on what the total project outflows are.

**Head is assuming minimum outflows and typical tailwater elevation of 2030 ft. Head at the dam is impacted by backwater from Box Canyon Dam, and high outflows can cause lower forebay elevations compared to lake elevations, which result in reduced head.



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EXISTING GATE INFORMATION

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Weight (Approximate, based on existing gate weight):

- Upper Leaf: ~47,000 lbs
- Lower Leaf: ~ 57,000 lbs

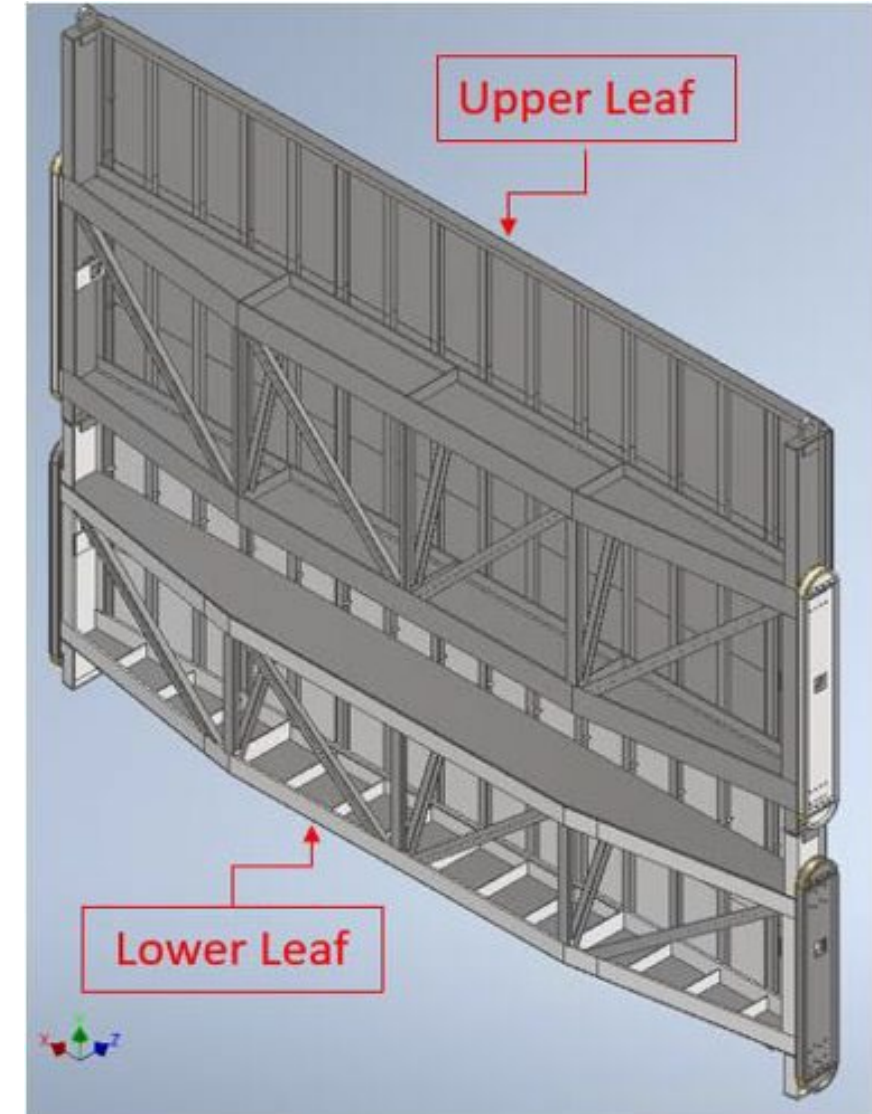
Dimensions: 41-feet wide x 32-feet high, split leaf vertical lift gate

- Upper Leaf: 41-feet wide by 19-feet high
- Lower Leaf: 41-feet wide by 13-feet high

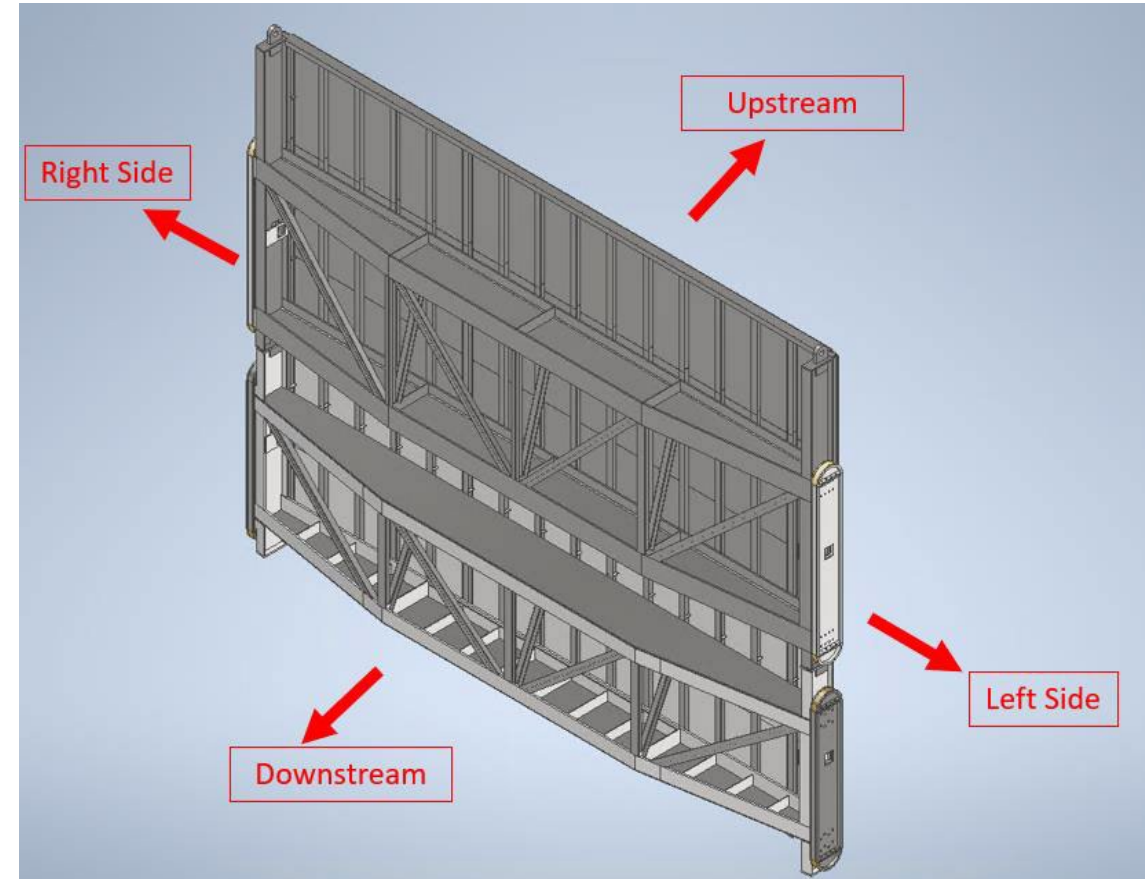
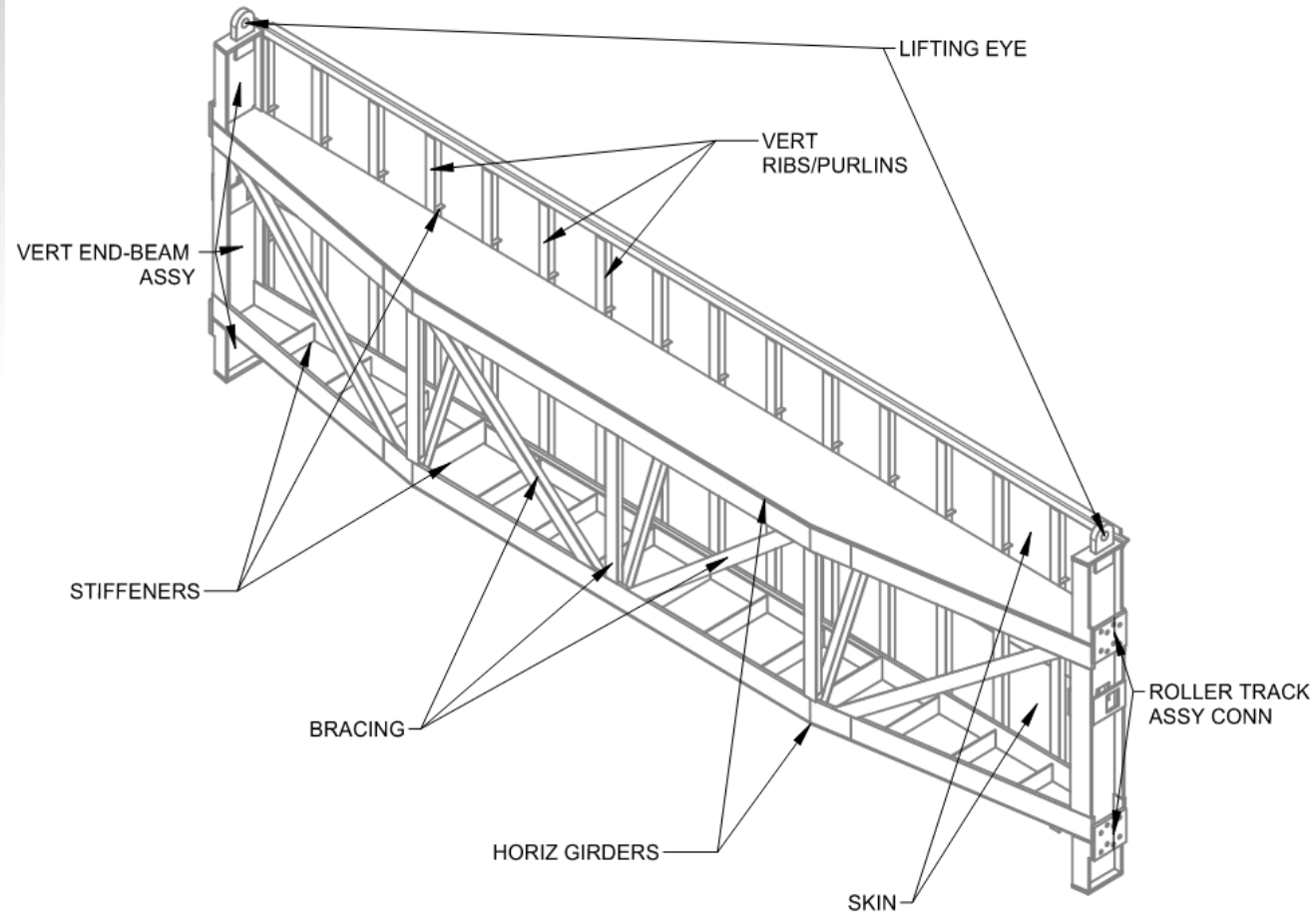
Quantity:

- Ten (10) total spillway gates
- One (1) spare gate for emergency and/or maintenance

Operated/lifted by gantry crane on the spillway deck.



EXISTING GATE CONFIGURATION





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PROJECT SCOPE

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Replacement of 3 gates, potential to award up to 11 gates total.

- Structural steel fabrication
- Non-destructive testing (NDT)
- Painting
- Seals
- Cathodic protection
- Lifting Beam
- Miscellaneous mechanical fabrication
- In-shop testing of various components
- Delivery with onsite crane and rigging
- Witness onsite fit and function test by Government
- Removal/Disposal of existing gate(s)

Timeline

- Award in FY2026
- First gate delivered to site in 2026 with lifting beam



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STRUCTURAL FABRICATOR REQUIREMENTS

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The work is performed according to but is not limited to the current versions AISC 325, **AWS D1.5M/D1.5**, and **AWS D1.6/D1.6** as well as the Unified Facilities Guide Specification (UFGS) SECTION 05 59 20 FABRICATION OF HYDRAULIC STEEL STRUCTURES.

Fabricating plants and fabricators may be required to have 5 years of documented experience on projects of similar scope and to be certified under the American Institute of Steel Construction (AISC) Quality Certification Program with one of the following certifications:

- Certified Bridge Fabricator - Intermediate (IBR)
- Certified Bridge Fabricator – Advanced (ABR)
- Certified Hydraulic Fabricator – Standard (HYDS)
- Certified Hydraulic Fabricator – Advanced (HYDA)

The fabricator will be required to hold a Fracture Control Endorsement (FCE) under the AISC Quality Certification Program, in addition to one of the certificates above.



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TECHNICAL QUALIFICATIONS FOR PAINTING

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- The painting contractor must be certified by the Association for Materials Protection and Performance (AAMP) with SSPC QP 3 certification for all shop surface preparation and coating application.
- Each painter will be required to get individually certified prior to painting. This is typically done via a 3rd party test agency or government witnessed test plate painting.
- The gates will be painted with 5-E-Z vinyl paint.



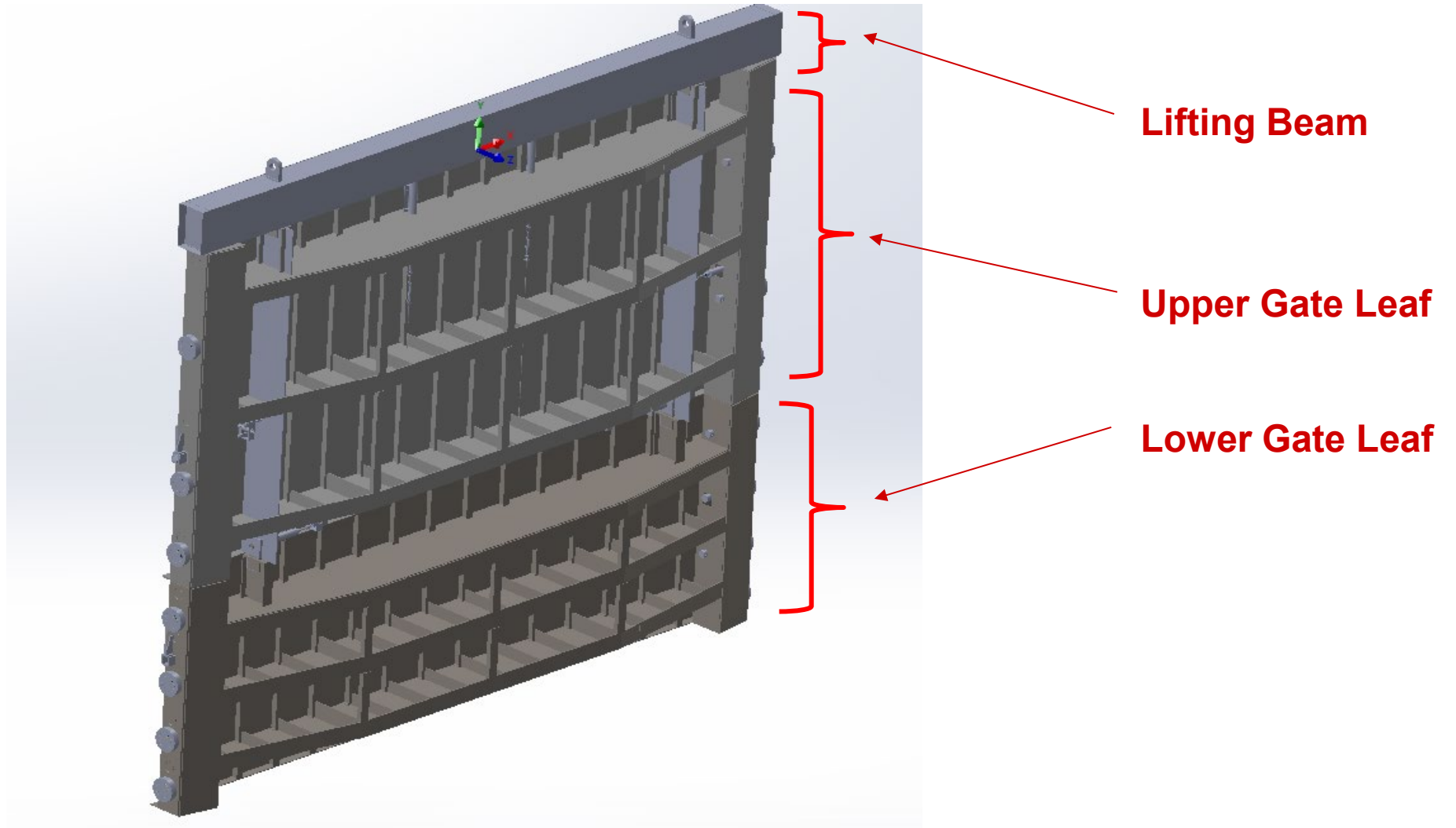
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CONCEPTUAL-STRUCTURAL

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- ASTM A-709 Grade 50 Steel required for the majority of new gate fabrication.





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CONCEPTUAL-MECHANICAL LINKAGES

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- Electric gear motors in lifting beam for actuation of latching and dogging mechanisms.
 - Latching of lifting beam to top leaf of gate.
 - Latching of top leaf to bottom leaf of gate
 - Dogging of gate in the gate slot.
- Mechanical linkages in top leaf of gate for latching and dogging
- Stainless steel wheels with spherical self-lubricated bearings in place of roller chain.





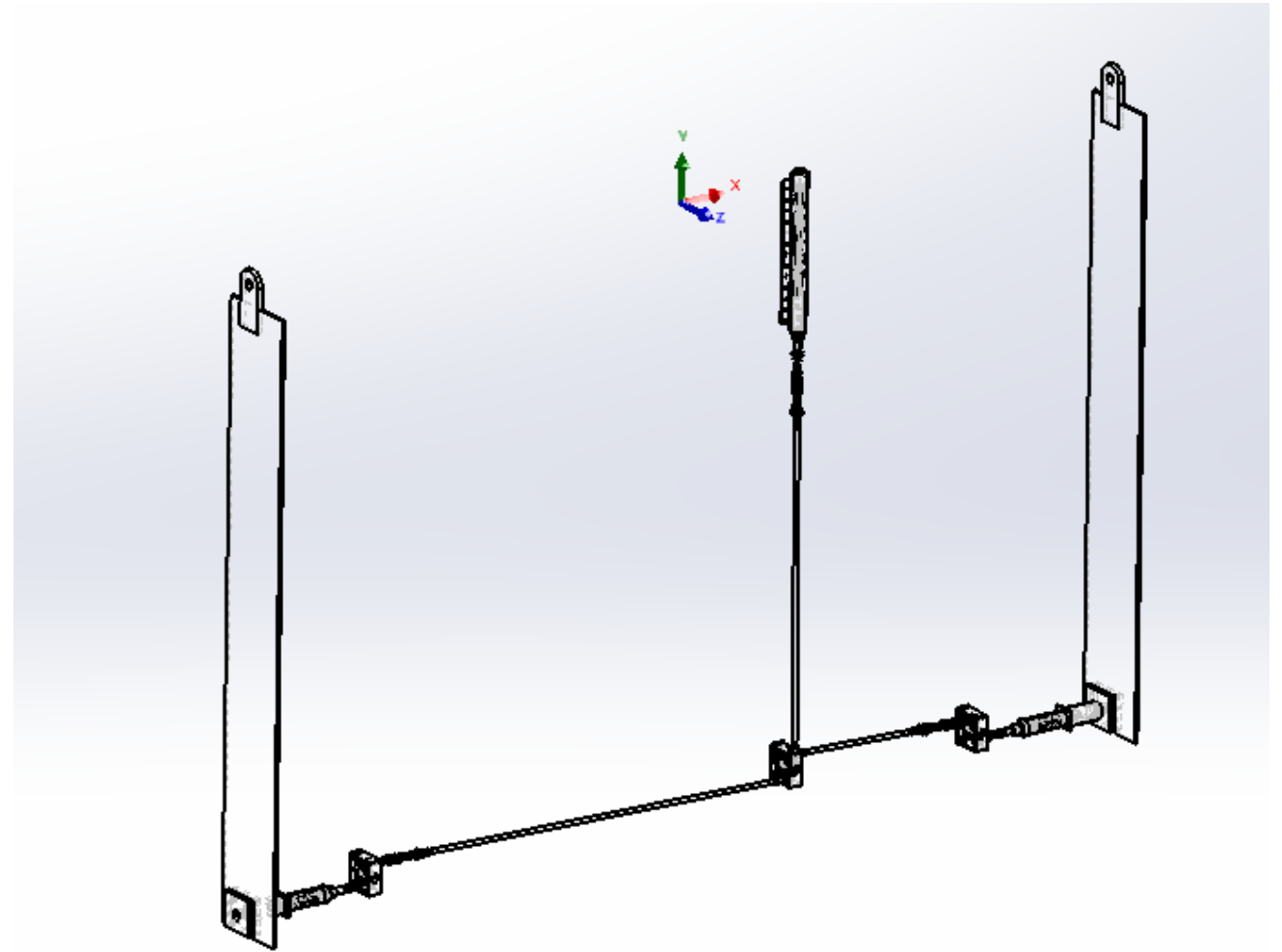
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CONCEPTUAL DESIGN – LATCHING MECHANISM

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- Springs for extension and retraction of latching pins.
- Actuated from lifting beam with a plunger.
- Stainless steel wire rope linkages.
- Bell cranks for change of direction.
- Dogging mechanism similar.





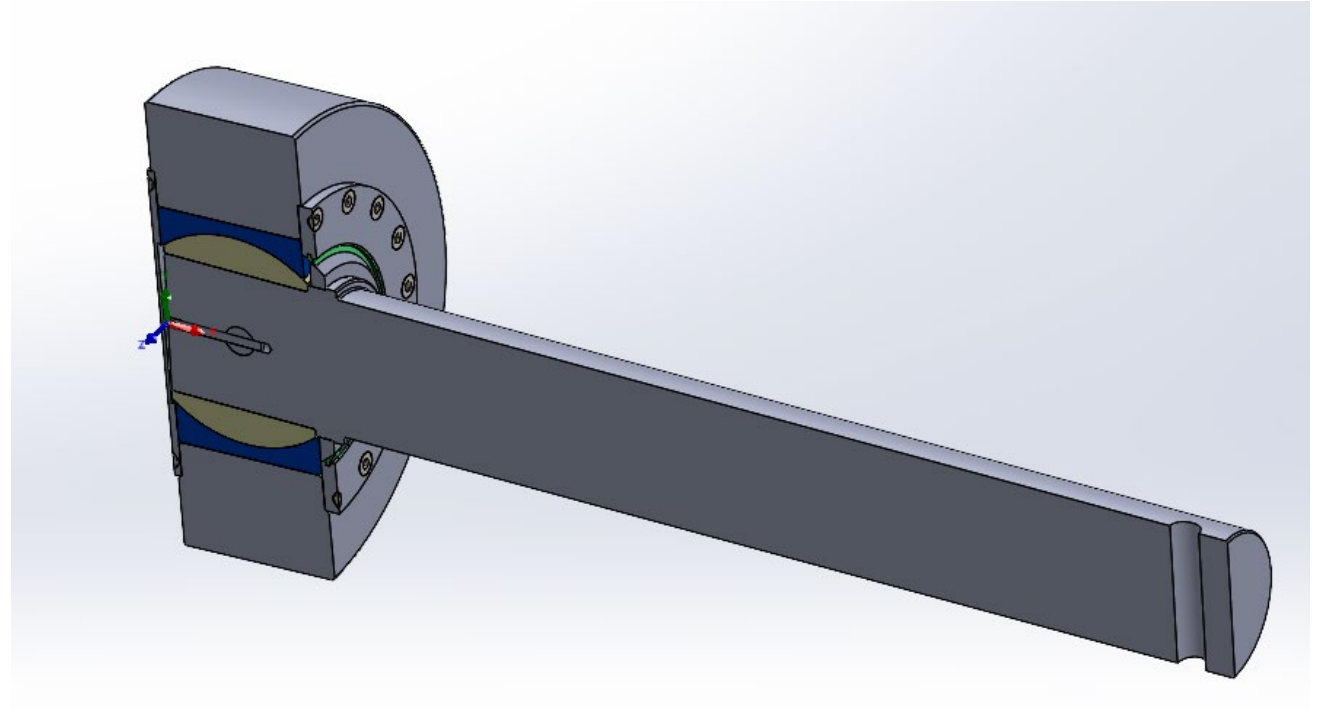
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CONCEPTUAL DESIGN - WHEELS

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- 16-inch Diameter by 5-inch Wide wheel
- Spherical self-lubricated bearing
- 17-4 PH Stainless steel rim, sphere and axle.
- Nitronic 60 fasteners





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CONCEPTUAL-DELIVERY ACCESS

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REMOVAL OF EXISTING GATES FOR DISPOSAL

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- Removal and disposal of gates may require remediation of hazardous paint and/or materials.
- Transition of existing gates for disposal expected to be the same transition locations for delivery of new gate(s).
- Removal of existing gate(s) would occur after replacement gate fit test is completed, or later as determined by Government based on ongoing and/or required operations.



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HOW TO STAY INFORMED?

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1. Fill out Attachment 2 of Industry Day posting and return by 4PM PST on February 6, 2025.
2. Check SAM.gov for updated information following event and for future notifications.
3. Any questions or comments, please feel free to contact Robert Gonzalez or Jeannette Patton.
4. Visit our website at: <https://www.nws.usace.army.mil/Missions/Civil-Works/Locks-and-Dams/Albeni-Falls-Dam/Spillway-Gate-Replacement-Project/>



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DIRECTIONS FOR ONSITE IF REGISTERED

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Address: Albeni Falls Dam, 2376 E Hwy 2, Oldtown, ID 83822



Albeni Falls Dam Visitor Center



Albeni Falls Dam Security Gate 1



Albeni Falls Dam Security Gate 2



Transition point towards Spillway Bridge





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QUESTIONS?